Course 7 overview

Hello, and welcome to Introduction to Data Analysis Using Python, the seventh course in the Google Data Analytics Certificate. You're on an exciting journey!

Throughout this course, you will develop an understanding of Python syntax, logic, data types, objects, and object-oriented programming. For many professionals, Python is the key to unlocking advanced analytics, machine learning, and the world of data science. By the end of this course, you will better understand how data analysts use programming on the job and how Python will be an important tool throughout your career as a data analytics professional.

1. Foundations: Data, Data, Everywhere
2. Ask Questions to Make Data-Driven Decisions
3. Prepare Data for Exploration
4. Process Data from Dirty to Clean
5. Analyze Data to Answer Questions
6. Share Data Through the Art of Visualization

7. Introduction to Data Analysis Using Python (this course)
8. Google Data Analytics Capstone: Complete a Case Study
9. Accelerate Your Job Search with AI

Course content

This course is divided into four modules. Here's an overview of the skills you'll learn in each module:

Module 1: Hello, Python!

You'll begin by exploring the basics of Python programming and why Python is such a powerful tool for data analysis. Python is an object-oriented programming language, and you'll discover the benefits of object-oriented programming for data professionals. You'll learn about Jupyter Notebooks, an interactive environment for coding and data work. And you'll investigate how to use variables and data types to store and organize your data.

Module 2: Functions and conditional statements

Next, you'll discover how to call functions to perform useful actions on your data. You'll also learn how to write conditional statements to tell the computer how to make decisions based on your instructions. And you'll practice writing clean code that can be easily understood and reused by other data professionals.

Module 3: Loops and strings

After that, you'll learn how to use iterative statements, or loops, to automate repetitive tasks. You'll also learn how to manipulate strings using slicing, indexing, and formatting.

Module 4: Data structures in Python

Then, you'll explore fundamental data structures such as lists, tuples, dictionaries, sets, and arrays. Lastly, you'll learn about two of the most widely used and important Python tools for advanced data analysis: NumPy and pandas.

What to expect

Each course offers many types of learning opportunities:

- Videos for instructors to teach new concepts and demonstrate the use of tools
- In-video questions that pop up from time to time to help you to check your understanding of key concepts and skills
- Step-by-step guides you can use to follow along with instructors as they demonstrate tools
- Readings to explore topics more in-depth and build on the concepts from the videos
- Connect with other learners: If you have a question, chances are, you're not alone. Visit Coursera's private Data Analytics Community to expand your network, discuss career journeys, and share experiences. Check out the quick start guide.
- Practice quizzes to prepare you for graded quizzes
- Graded quizzes to measure your progress and give you valuable feedback

This program was designed to let you work at your own pace—your personalized deadlines are just a guide. There is no penalty for late assignments. To earn your certificate, you simply need to complete all of the work.

If you miss two assessment deadlines in a row, or if you miss an assessment deadline by two weeks, you'll see a Reset deadlines option on the Grades page. Click it to switch to a new course schedule with updated deadlines. You can use this option as many times as you need—it won't remove any progress you've already made in the course, but you may find new course content if the instructor updated the course after you started. If you cancel a subscription and then reactivate it, your deadlines will automatically reset.

In this course, you'll be assessed with graded quizzes and activities. Both are based on the wide variety of learning materials and activities that reinforce the important skills you'll develop. And both can be taken more than once.

Tips for success

- It is strongly recommended that you go through the items in each lesson in the order they appear because new information and concepts build on previous knowledge.
- Participate in all learning opportunities to gain as much knowledge and experience as possible.
- If something is confusing, don't hesitate to replay a video, review a reading, or repeat a self-review activity.
- Use the additional resources that are referenced in this course. They are designed to support your learning. You can find all of these resources in the Resources tab.
- When you encounter useful links in this course, bookmark them so you can refer to the information later for study or review.

Understand and follow the Coursera Code of Conduct to ensure that the learning community remains a welcoming, friendly, and supportive place for all members.